

MA 301
Final
Fall - 2013

Problem	points	out of
1		8
2		8
3		8
4		8
5		8
6		16
7		8
8		22
Score		86

Name: _____

HR: _____

Signature: _____

READ BEFORE YOU START
* Show all work. * Change your calculator mode to RADIANS. * Sloppy solutions may be penalized. * Use calculators ONLY for computations. * You must write up every problem in a clear and organized way. * If you write solutions on a page different than the question, then clearly state where it is located.

Problem 1: Given $\mathbf{F} = [e^x \cos y + yz, xz - e^x \sin y, xy + z]$

(a) Show that $\mathbf{F}$ is conservative

(b) Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ when C is a curve from $(0, 0, 2)$ to $\left(0, \frac{3\pi}{2}, -6\right)$

Problem 2: Let $\mathbf{F} = [3z^2, 6, 6xz]$ and S is the parabolic cylinder $S : y = x^2$ where $0 \leq x \leq 2$ and $0 \leq z \leq 3$ and normal vector point away from the y -axis. Compute the flux of $\mathbf{F}$ through the parabolic cylinder S

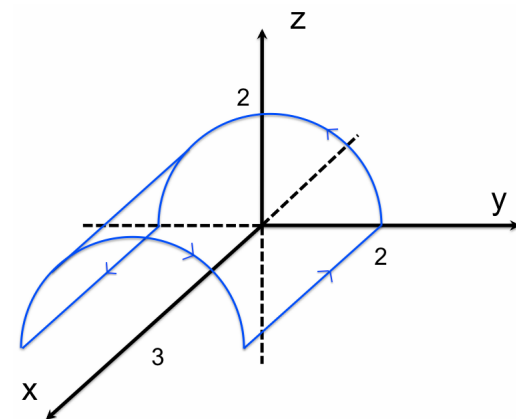
Problem 3: Let $\mathbf{F} = [16xy + 3yx^2, x^3 + 2y]$ and let C be a boundary of the region bounded by: $x = 3$, $y = 0$, and $y = x$.

Compute $\oint_C \mathbf{F} \cdot d\mathbf{r}$

Problem 4: Evaluate $\iint_S \mathbf{F} \cdot \mathbf{n} \, dA$ if $F = [x^3 - y^3, y^3 - z^3, z^3 - x^3]$ and S is given by: $x^2 + y^2 + z^2 \leq 25$, $z \geq 0$ with outward pointing normal.

Problem 5: Let $\mathbf{F} = [-y, 2z, 0]$ and C be the boundary of side of the half cylinder $y^2 + z^2 = 4$, with $z \geq 0$, $0 \leq x \leq 3$ and oriented outward (see below).

Use Stokes' thm to compute $\int_C \mathbf{F} \cdot d\mathbf{r}$.



Problem 6: Let $z_1 = -2 - 2i$, $z_2 = i$. Answer the following and write answers in the specified form:

(a) (*Standard Form*) $z_1 z_2 =$

(b) (*Standard Form*) $z_1 - z_1 z_2 =$

(c) $|z_1 z_2| =$

(d) Convert z_1 to Euler's Polar form:

(e) Convert z_2 to Euler's Polar form:

(f) (*Euler's Polar Form*) $(z_2)^{76} =$

(g) (*Euler's Polar Form*) Solve for z : $\frac{1}{2}z^3 - z_1 = 0$ (explicitly write out all solutions)

Problem 7: Let $f(x) = \begin{cases} \pi & -\frac{\pi}{2} < x < 0 \\ -\pi & 0 < x < \frac{\pi}{2} \end{cases}$ and $f(x) = f(x + \pi)$

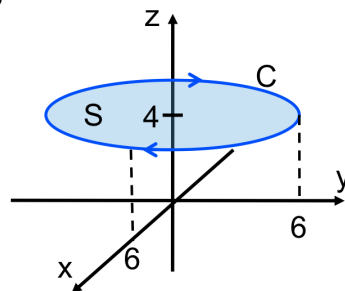
Find the Fourier Series of $f(x)$

Problem 8: Answer the following true/false and short-answer questions:

T / F	The line integral of any vector field $\mathbf{F}$ around any closed curve C is zero
T / F	Let C be an ellipse. If $\int_C \mathbf{F} \cdot d\mathbf{r} = 0$, then $\mathbf{F}$ is path-independent.
T / F	If $\mathbf{F}$ is path-independent, then there is a potential function for $\mathbf{F}$
T / F	If $\mathbf{F} = \nabla f$ then $\mathbf{F}$ is conservative.
T / F	If $\mathbf{F}$ is path-independent and C is any closed curve, then $\int_C \mathbf{F} \cdot d\mathbf{r} = 0$
T / F	If S_1 is a rectangle with area = 1 and S_2 is a rectangle with area = 2, then $2 \iint_{S_1} \mathbf{F} \cdot d\mathbf{A} = \iint_{S_2} \mathbf{F} \cdot d\mathbf{A}$
T / F	Suppose $\mathbf{F} = [1, 1, 1]$ and S is a closed surface with positive orientation. Then $\iint_S \mathbf{F} \cdot \mathbf{n} dA > 0$.
T / F	If $\mathbf{F}$ is a vector-field such that $\text{div} \mathbf{F} = 3$ and S is a closed surface oriented outward, then $\iint_S \mathbf{F} \cdot d\mathbf{S}$ is equal to one-third the volume enclosed by S .
T / F	Suppose $f(x) = f(x + 6)$ for all x , then we say f is periodic with period 3.
T / F	$\sqrt{-5} \cdot \sqrt{-5} = 5$.

Let $\mathbf{F}(x, y, z) = [zy - 3y, xz - z, xy + 1]$ with C and S given in the figure.

$$\int_C \mathbf{F} \cdot d\mathbf{r} =$$



Let $\mathbf{F} = \nabla(x^2 \sin y)$ and let C be the boundary of region between: $x = 1$, $y = 0$, and $y = \sqrt{x}$ oriented counter-clockwise

$$\oint_C \mathbf{F} \cdot d\mathbf{r} =$$

Let $\mathbf{F} = [6x, ye^z, -e^z]$ and S be the surface (with outward orientation) of the box $|x| \leq 1$, $|y| \leq 2$, $0 \leq z \leq 1$.

$$\iint_S \mathbf{F} \cdot \mathbf{n} dA =$$